

MATH 3115 MSc & SED COMPLEX ANALYSIS

Lecture 12: RESIDUE THEORY

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Lesson Outcome

By the end of this lesson you should be able to:

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TAYLOR SERIES

INTRODUCTION

- We have seen that if f is analytic at z_0 , then the derivatives $f^{(n)}(z_0)$ exist for all n .
- In this section, we will see that if f is analytic at z_0 , then f has a complex Taylor series representation centered at z_0 .

TAYLOR SERIES

Definition 1: Taylor Series

If $f(z)$ is analytic at z_0 , then the series

$$f(z) = f(z_0) + f'(z_0)(z - z_0) + \frac{f^{(2)}(z_0)}{2!}(z - z_0)^2 + \frac{f^{(3)}(z_0)(z - z_0)^3}{3!} \\ + \cdots = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$$

is called the Taylor series for f centered at z_0

TAYLOR SERIES CONT'D

Definition 2: Maclaurin series

If $f(z)$ is analytic at $z_0 = 0$, then the series

$$f(z) = f(0) + f'(0)z + \frac{f^{(2)}(0)}{2!}z^2 + \frac{f^{(3)}(0)}{3!}z^3 + \dots$$

$= \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} z^n$ is called the Maclaurin Series for f centered at 0 .

Taylor's Theorem

If f is analytic in the simply connected domain D that contains the circle $|z - z_0| = R$, then the Taylor Series in definition 1 converge to $f(z)$ in the disk $|z - z_0| < R$; that is $f(z) = \sum_{n=0}^{\infty} \frac{f^{(n)}(z_0)}{n!} (z - z_0)^n$ valid for $|z - z_0| < R$

TAYLOR SERIES CONT'D

Example 1

Find a Taylor Series representation of $f(z) = \sin z$ centered at $z = \pi/2$.

Solution to Example 1

$$f(z) = \sin z, \quad z_0 = \frac{\pi}{2}, \quad f\left(\frac{\pi}{2}\right) = \sin \frac{\pi}{2} = 1$$

$$f'(z) = \cos z, \quad f'\left(\frac{\pi}{2}\right) = \cos \pi/2 = 0$$

$$f''(z) = -\sin z, \quad f''\left(\frac{\pi}{2}\right) = -\sin \pi/2 = -1$$

$$f''(z) = -\cos z, \quad f'''\left(\frac{\pi}{2}\right) = -\cos \frac{\pi}{2} = 0$$

$$f^{(4)}(z) = \sin z, \quad f^{(4)}\left(\frac{\pi}{2}\right) = \sin \pi/2 = 1$$

Solution to Example 1 CONT'D

$$f(z) = f(\pi/2) + \frac{f'(\pi/2)}{1!}(z - \pi/2) + \frac{f''(\pi/2)}{2!}(z - \pi/2)^2 + \frac{f'''(\pi/2)(z - \pi/2)}{3!} \\ + \frac{f^{(4)}(\pi/2)}{4!}(z - \pi/2)^4 + \dots$$

$$= 1 + \frac{0}{1!}(z - \pi/2) + \frac{(-1)}{2!}\left(z - \frac{\pi}{2}\right)^2 + \frac{0}{3!}(z - \pi/2)^3 + \frac{(11)(z - \pi/4)}{4!}$$

$$= 1 - \frac{(z - \frac{\pi}{2})^2}{2!} + \frac{(z - \frac{\pi}{2})^4}{4!} - \frac{(z - \frac{\pi}{2})^6}{6!}$$

$$f(z) = \sum_{n=0}^{\infty} \frac{(-1)^{2n}}{n!} \left(z - \frac{\pi}{2}\right)^n$$

Example 2

Find a Taylor series expansion of $f(z) = \frac{1}{z}$ centered at $z = 1$

Taylor series

Solution to Example 2

$$f(z) = 1/2, f(1) = 1/1 = 1$$

$$f'(2) = -1/2^2, f'(1) = -\frac{1}{1} = -1$$

$$f''(z) = +2/2^3, f''(1) = \frac{2}{1^3} = 2$$

$$f'''(z) = -6/\pi^4, f'''(1) = \frac{-6}{1^4} = -6$$

$$f^{(4)}(z) = 24/\pi^5, f^{(4)}(1) = 24$$

$$f(z) = f(1) + \frac{f'(1)}{1!}(z-1)^1 + \frac{f''(1)}{2!}(z-1)^2 + \frac{f'''(1)}{3!}(z-1)^3 + \frac{f^{(4)}(1)}{4!}(z-1)^4$$

+ ...

$$= 1 + \frac{(-1)}{1!}(z-1) + \frac{2(z-1)^2}{2!} - \frac{6(z-1)^3}{3!} + \frac{24(z-1)^4}{4!}$$

$$= 1 - (z-1)^1 + (z-1)^2 - (z-1)^3 + (z-1)^4 + \dots$$

$$= \sum_{n=0}^{\infty} (-1)^n (z-1)^n$$

MACLAURIN SERIES EXAMPLE

Example 3

Find the Maclaurin series for $\log(1 + z)$ centered at $z = 0$.

Solution 3

$$f(z) = \log(1+z) \quad f(0) = 0$$

$$f'(z) = \frac{1}{1+z}, \quad f'(0) = 1$$

$$f''(z) = \frac{-1}{(1+z)^2}; \quad f''(0) = -1$$

$$f'''(z) = 2/(1+z)^3, \quad f'''(0) = 2$$

$$f^{(4)}(z) = -6/(1+z)^4, \quad f^{(4)}(0) = -6$$

$$\begin{aligned} f(z) &= f(0) + \frac{f'(0)}{1!}z + \frac{f''(0)}{2!}z^2 + \frac{f'''(0)}{3!}z^3 + \frac{f^{(4)}(0)}{4!}z^4 + \cdots \\ &= 0 + \frac{1}{1!}z + \frac{(-1)z^2}{2!} + \frac{2z^3}{3!} + \frac{(-6)z^4}{4!} + \cdots \\ &= z - \frac{z^2}{2} + \frac{z^3}{3} - \frac{z^4}{4} + \cdots \\ &= \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{n} z^n \end{aligned}$$

LAURENT SERIES

Introduction

- Suppose that f is analytic in a deleted neighborhood $0 < |z| < R$, but it is not analytic at $z = 0$.
- Then f will not have a Maclaurin series expansion of f .
- For example $f(z) = \frac{\sin z}{z^2}$ is analytic for all $z \neq 0$.
- However if we use the Maclaurin series for $\sin z$ and formally divide each term by z^2 , we get:

$$\begin{aligned} f(z) &= z^{-2} \sin z = z^{-2} \left(z - \frac{z^3}{3!} + \frac{z^5}{5!} - \frac{z^7}{7!} + \cdots \right) \\ &= \frac{z}{z^2} - \frac{z^3}{3!z^2} + \frac{z^5}{5!z^2} - \frac{z^7}{7!z^2} + \cdots \\ &= \frac{1}{z} - \frac{z}{6} + \frac{z^3}{120} - \frac{z^5}{5040} + \cdots \quad \text{for } |z| > 0 \end{aligned}$$

LAURENT SERIES CONT'D

- The representation from the above slide is an example of Laurent series centered at z_0
- This example generalizes the Taylor series method by permitting negative powers of z to be used in the representation.

LAURENT SERIES CONT'D

Theorem

Let $r \geq 0$ and $R > r$, and suppose that f is analytic in the annulus $A = \{z : r < |z - z_0| < R\}$ which is shown in the figure below. Then f has the Laurent series representation

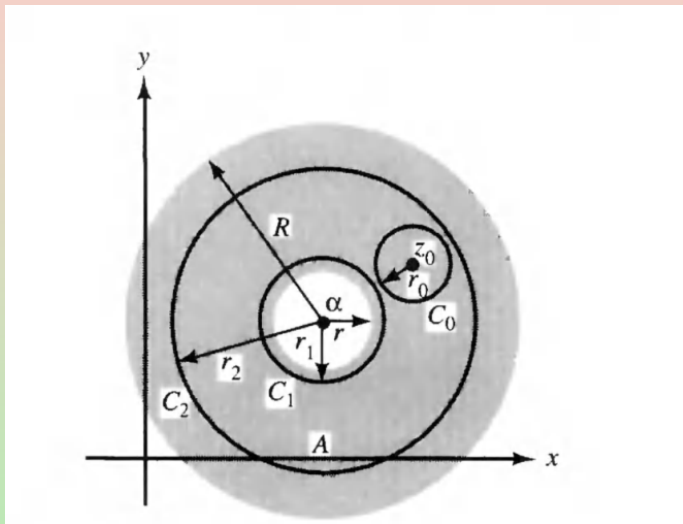
$$f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n \text{ valid for } r < |z - z_0| < R$$

Where the Laurent series coefficients a_n are given by

$$a_n = \frac{1}{2\pi i} \int_C \frac{f(\varepsilon) d\varepsilon}{(\varepsilon - z_0)^{n+1}} \text{ for } n = 0, \pm 1, \pm 2, \pm 3, \dots$$

where C is any positively oriented circle with center z_0 and radius ρ where $r < \rho < R$.

LAURENT SERIES CONT'D



- The circles C_0, C_1, C_2 lie in the annulus A .

Example 4

Find three different Laurent series expansion for $f(z) = \frac{3}{2+z-z^2}$ involving powers of z .

Solution 4

- The function f has poles at $z = -1, 2$,
- And is analytic in the disk $D : |z| < 1$, in the annulus $A : 1 < |z| < 2$, and in the region $R : |z| > 2$.
- We will find a different Laurent series for f in each of the three domains D , A , and R .
- We perform partial fraction decomposition

SOLUTION TO EXAMPLE 4 CONT'D

$$f(z) = \frac{3}{(1+z)(2-z)} = \frac{1}{1+z} + \frac{1}{2-z} = \frac{1}{1+z} + \frac{1}{2} \frac{1}{1-\left(\frac{z}{2}\right)}$$

- For $\frac{1}{1+z}$, the expansion valid for $|z| < 1$ is:

$$\frac{1}{1+z} = \sum_{n=0}^{\infty} (-1)^n z^n$$

- For $\frac{1}{1-\frac{z}{2}}$, (which is geometric series formulae) the expansion valid $|\frac{z}{2}| < 1 \implies |z| < 2$

$$\frac{1}{2} \times \left(\frac{1}{1-\frac{z}{2}} \right) = \frac{1}{2} \times \sum_{n=0}^{\infty} \left(\frac{z}{2} \right)^n = \frac{1}{2} \times \sum_{n=0}^{\infty} \frac{z^n}{2^n} = \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}$$

- Combining the expansions for the two series $\frac{1}{1+z}$ and $\frac{1}{2-z}$ when $|z| < 1$

$$f(z) = \left(\sum_{n=0}^{\infty} (-1)^n z^n \right) + \left(\sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} \right).$$

$$f(z) = \sum_{n=0}^{\infty} \left[(-1)^n + \frac{1}{2^{n+1}} \right] z^n.$$

Solution 4 CONT'D

- Consider the series expansion valid for $1 < |z| < 2$,
- For $\frac{1}{1+z}$, the expansion valid for $|z| > 1$:

$$\frac{1}{1+z} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{z^n}$$

- We know that $\frac{1}{2} \frac{1}{(1-\frac{z}{2})} = \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}}$ valid for $|z| < 2$

- Combining the two we have

$$f(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{z^n} + \sum_{n=0}^{\infty} \frac{z^n}{2^{n+1}} \text{ for } 1 < |z| < 2$$

- For $\frac{1}{1+z} = \frac{1}{z} \sum_{n=0}^{\infty} \frac{(-1)^n}{z^n} = \sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{z^n}$ valid $|z| > 2$

Similarly, For $\frac{1}{2} \frac{1}{(1-\frac{z}{2})}$ the expansion valid for $|z| > 1$

$$\frac{\frac{1}{2}}{1-\left(\frac{z}{2}\right)} = \sum_{n=1}^{\infty} \frac{-2^{n-1}}{2^n} \text{ valid for } |z| > 2$$

- Hence $f(z) = \sum_{n=1}^{\infty} \frac{(-1)^{n+1} - 2^{n-1}}{z^n}$ valid $|z| > 2$.

Example 5

Find the Laurent series representation $f(z) = \frac{(\cos z - 1)}{z^4}$ that involves powers in z .

Solution to Example 5

We can use the Maclaurin series for $\cos z - 1$

$$\begin{aligned}\cos z &= 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots \\ \cos z - 1 &= 1 - \frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots - 1 \\ &= -\frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots \\ f(z) &= \frac{-\frac{z^2}{2!} + \frac{z^4}{4!} - \frac{z^6}{6!} + \cdots}{z^4} \\ &= \frac{-1}{2z^2} + \frac{1}{24} - \frac{z^2}{720} + \cdots\end{aligned}$$

valid for $z \neq 0$.

PRINCIPAL PART OF LAURENT SERIES

Definition

The principal part of a Laurent series is the series of terms with negative degree (powers), that is

$$\sum_{k=-\infty}^{-1} a_k (z - c)^k$$

PRINCIPAL PART OF LAURENT SERIES

- To find the principal part of a function, identify the terms in the Laurent series that correspond to the negative powers of z because the principal part of a function consists of all terms involving negative powers of z
- If the principal part of f is a finite sum, then f has a pole at c of order equal to (negative) the degree of the highest term;
- On the other hand, if f has an essential singularity at c , the principal part is an infinite sum (meaning it has infinitely many non-zero terms).

PRINCIPAL PART OF LAURENT SERIES

Example 6

Consider the function:

$$f(z) = \frac{1}{z^2} + \frac{3}{z} + z + 1.$$

Find the principal part of this function around $z = 0$.

PRINCIPAL PART OF LAURENT SERIES

Solution to Example 6

The function is already written in a form resembling a Laurent series:

$$f(z) = \frac{1}{z^2} + \frac{3}{z} + z + 1$$

So by Laurent series around $z = 0$ we:

$$f(z) = \sum_{n=-\infty}^{\infty} c_n z^n = \frac{1}{z^2} + \frac{3}{z} + z + 1$$

The term $\frac{1}{z^2}$ has a negative power of z . Thus z^{-2} .

The term $\frac{3}{z}$ also has a negative power of z . Thus $3z^{-1}$.

From the given function we see that, the terms with negative power are:

$\frac{1}{z^2}$ and $\frac{3}{z}$

Hence, the principal part of $f(z)$ is: $\frac{1}{z^2} + \frac{3}{z}$

RESIDUES

Introduction

- The Cauchy Integral Formulae are useful in evaluating contour integrals over a simple closed contour C where the integrand has the form $f(z)/(z - z_0)^k$ and f is an analytic function.
- We extend this result to integrals that have a finite number of singularities that lie inside the contour Γ .

RESIDUE CONTINUED

Definition

Let f have a non-removable isolated singularity at the point z_0 . Then f has a Laurent series expansion $f(z) = \sum_{n=-\infty}^{\infty} a_n (z - z_0)^n$ valid for $0 < |z - z_0| < 1$. The coefficient a_{-1} of $\frac{1}{(z - z_0)}$ is called the residue of f at z_0 , and we use the notation $\text{Res}(f, z_0) = a_{-1}$

RESIDUE CONT'D

Example 6

Find the residue of $f(z) = e^{1/z}$ at $z = 0$.

Solution 6

$f(z)$ has a Laurent expansion

$$f(z) = e^z = 1 + z + \frac{z^2}{2!} + \frac{z^3}{3!} + \dots$$

$$f(z) = e^{2/z} = 1 + \frac{2}{z} + \frac{2^2}{2!z^2} + \frac{2^3}{3!z^3} + \dots$$

The coefficient of $\frac{1}{z}$ is 2 The $\text{res}(f, 0) = 2$

RESIDUE CONT'D

Example 7

Compute the residue of the function $f(z) = z^2 \sin\left(\frac{1}{z}\right)$ at $z = 0$.

Solution to Example 7

The function $f(z) = z^2 \sin\left(\frac{1}{z}\right)$ has a Laurent series expansion:

$$\begin{aligned}\sin\left(\frac{1}{z}\right) &= \frac{1}{z} - \frac{1}{3!z^3} + \frac{1}{5!z^5} - \frac{1}{7!z^7} + \cdots \\ z^2 \sin\left(\frac{1}{z}\right) &= z - \frac{1}{3!z} + \frac{1}{5!z^3} - \frac{1}{7!z^5} + \cdots\end{aligned}$$

The coefficient of $\frac{1}{z}$ is $\frac{1}{3!} = \frac{1}{6}$. The residue of f at $z = 0$ is $\frac{1}{6}$.

CAUCHY RESIDUE THEOREM

Theorem

Let D be a simply connected domain, and let C be a simply closed positively oriented contour that lies in D . If f is analytic on C and inside C except at the points $z_1, z_2, \dots, z_n$ that lie inside C , then

$$\int_C f(z) dz = 2\pi i \sum_{k=1}^n \text{Res}(f, z_k)$$

CAUCHY RESIDUE THEOREM CONT'D

Example 8

Show that $\int_C \exp(2/z) = 4\pi i$ where C is the unit circle $|z| = 1$ with positive orientation

Solution

$f(z) = \exp\left(\frac{2}{z}\right)$ at $z_0 = 0$ is $\text{Res}(f, 0) = 2$. So

$$\int_C \exp(2/z) dz = 2\pi i R(f, 0) = 2\pi i(2) = 4\pi i$$

- The exponential function is denoted as $\exp\left(\frac{2}{z}\right) = e^{\frac{2}{z}}$

RESIDUES AT POLE

- The calculation of a Laurent series expansion is tedious in most cases.
- So we seek a method to calculate the residue from special information about the nature of the singularity at z_0 .

Definition

If f has a removable singularity at z_0 , then $a_{-n} = 0$ for $n = 1, 2, \dots$. Therefore if z_0 is a removable singularity, then $\text{Res}(f_1 z_0) = 0$

RESIDUE AT POLE CONT'D

Theorem: Residue at Pole

(i) If f has a simple pole at z_0 , then

$$\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} (z - z_0) f(z)$$

(ii) If f has a pole of order 2 at z_0 , then

$$\text{Res}(f, z_0) = \lim_{z \rightarrow z_0} \frac{d}{dz} (z - z_0)^2 f(z)$$

(iii) If f has a pole of order k at z_0 , then

$$\text{Res}(f, z_0) = \frac{1}{(k-1)!} \lim_{z \rightarrow z_0} \frac{d^{k-1}}{dz^{k-1}} (z - z_0)^k f(z)$$

Example 8

Find $\int_c \frac{dz}{(z^4+z^3-2z^2)}$ where c is the circle

Solution to Example 8

So we $f(z) = \frac{1}{(z^4+z^3-2z^2)} = \frac{1}{z^2(z^2+z-2)} = \frac{1}{z^2(z+2)(z-1)}$

The singularities of f all lie inside C , $|z| = 3$.

$z = -2$ is a pole of order 1 (simple pole)

$z = 0$ is a pole of order 2

$z = 1$ is a pole of order 1 (simple pole)

So we compute the residues as follows:

$$\begin{aligned}\operatorname{Res}(f, -2) &= \lim_{z \rightarrow -2} (z+2)f(z) = \lim_{z \rightarrow -2} (z+2) \frac{1}{z^2(z+2)(z-1)} \\ &= \lim_{z \rightarrow -2} \frac{1}{z^2(z-1)} = \frac{1}{(-2)^2(-2-1)} = \frac{1}{4(-3)} = -\frac{1}{12}\end{aligned}$$

Solution to Example 8 CONT'D

$$\begin{aligned}\operatorname{Res}(f, 0) &= \lim_{z \rightarrow 0} \frac{d}{dz} z^2 \left(\frac{1}{z^2(z+2)(z-1)} \right) \\&= \lim_{z \rightarrow 0} \frac{d}{dz} \left(\frac{1}{z^2 + z - 2} \right) \\&= \lim_{z \rightarrow 0} \frac{(1)'(z^2 + z - 2) - (z^2 + z - 2)'(1)}{(z^2 + z - 2)^2} = \lim_{z \rightarrow 0} \frac{-(2z + 1)}{(z^2 + z - 2)^2} \\&= -\frac{1}{4}\end{aligned}$$

$$\operatorname{Res}(f(1)) = \lim_{z \rightarrow +1} (z-1) \frac{1}{z^2(z+2)(z-1)} = \lim_{z \rightarrow +1} \frac{1}{z^2(z+2)} = \frac{1}{3}$$

By using the Residue Theorem

$$\int \frac{dz}{z^4 + z^3 - 2z^2} = 2\pi i \left(-\frac{1}{4} + \frac{1}{3} - \frac{1}{12} \right) = 0.$$

Example 9

Decompose $f(z) = \frac{z^2+3z+2}{z^2(z-1)}$ into partial fractions.

Solution to Example 9

$$f(z) = \frac{z^2+3z+2}{z^2(z-1)} = \frac{A}{z^2} + \frac{B}{z} + \frac{C}{z-1}$$

$$\text{Res}(z^2 f(z), 0) = \lim_{z \rightarrow 0} \frac{z^2 (z^2 + 3z + 2)}{z^2(z-1)} = \lim_{z \rightarrow 0} \frac{z^2 + 3z + 2}{z-1} = -2$$

$$\begin{aligned} \text{Res}(zf(z), 0) &= \lim_{z \rightarrow 0} \frac{d}{dz} \frac{z^2 (z^2 + 3z + 2)}{z^2(z-1)} = \lim_{z \rightarrow 0} \frac{d}{dz} \left(\frac{z^2 + 3z + 2}{z-1} \right) \\ &= \lim_{z \rightarrow 0} \frac{(2z+3)(z-1) - (z^2+3z+2)}{(z-1)^2} = -5 \end{aligned}$$

$$\text{Res}(f, 1) = \lim_{z \rightarrow 1} \frac{z^2 + 3z + 2}{z^2} = 6$$

$$\text{So } \frac{z^2 + 3z + 2}{z^2(z-1)} = \frac{-2}{z^2} - \frac{5}{z} + \frac{6}{z-1}$$

READING ASSIGNMENT : IMPROPER INTEGRALS IN COMPLEX PLANE

THE END